

SE3010: Software Engineering Process & Quality Management

Year 3 Semester 1

2026

Assignment Title	Assignment 1
Learning outcomes covered	LO 1: LO 2:
Assignment Mode	Online
Maximum Marks	100%
Contribution to the Final Grade	20%
Date published	2026/02/21
Deadline for submissions	2026/03/07
Mode of Submission	Online

Assignment description

Part 1: Tool Selection

Each group must select **one software testing tool** from the list below:

- PyTest, TestNG, Cucumber, Cypress, Playwright, Robot Framework, Karate, JMeter, Gatling, NUnit, Mocha / Chai, Vitest, RSpec, Behave, k6, Locust

Selenium, JUnit and Postman are strictly not allowed.

Part 2: Each group must identify one industry professional currently working in a software company.

The industry professional **must have hands-on, practical experience with the same software testing tool selected by the group.**

Industry Interview – Discussion Areas

The interview discussion should focus on how the selected tool contributes to Quality Engineering practices across the software development lifecycle, rather than on testing alone.

A. Background in Quality Engineering

- Briefly describe your role and responsibilities related to Quality Engineering within your organization.
- How long have you been involved in software quality, testing, or quality engineering?
- What types of projects do you primarily work on (e.g., web, mobile, enterprise, cloud, distributed systems)?

B. Quality Engineering Tool Experience and Context

- How is the selected tool used to ensure and improve software quality in real projects?
- How long have you been using this tool as part of your quality engineering practices?
- At which stages of the software development lifecycle is this tool mainly used (e.g., design, development, CI/CD, post-deployment)?

C. Quality Engineering Practices Using the Tool

- Which features of the selected tool contribute most to quality assurance and defect prevention?
- What types of quality checks are implemented using this tool (e.g., unit quality, integration quality, system reliability, performance quality)?
- How are test cases or quality checks structured, reviewed, and maintained to ensure long-term quality?

D. Quality Risks and Automation Challenges

- What quality-related challenges arise when using this tool in real projects?
- How do you handle quality risks such as flaky tests, unstable pipelines, or unreliable test results?
- In which situations is human-driven quality validation preferred over automation, and why?

E. Quality Engineering in CI/CD and DevOps

- How is the selected tool integrated into the CI/CD pipeline as part of quality gates?
- How frequently are automated quality checks executed, and at what stages?
- How are quality metrics, test results, and failures communicated to developers and stakeholders?

F. Skills, Expectations, and Professional Advice in Quality Engineering

- What level of quality engineering or test automation expertise do you expect from fresh graduates?
- Which skills should students focus on to become effective Quality Engineers rather than only testers?

- What common quality-related mistakes do junior engineers make when adopting automation tools?
- How does academic teaching of software testing differ from industry-level quality engineering practices?
- What advice would you give students aiming to pursue a career in Quality Engineering or Test Automation?

Part 3: Presentation Requirements

Each group must deliver a short presentation covering the following aspects:

- Introduction to the selected tool
- Justification for selecting the tool
- Key features of the tool
- Industry adoption and how widely the tool is used

The presentation should be 10 minutes maximum.

Part 4: Demonstration Requirements

Each group must demonstrate the **complete workflow of using the selected testing tool**, covering the process from **initial setup** to successful **test execution**.

- Each member needs to demonstrate **at least 1 key feature (Per group 4 features)** of the tool, such as:
 - Assertions
 - Fixtures or setup/teardown mechanisms
 - Mocking or stubbing
 - Behaviour-Driven Development (BDD) syntax
 - Test reporting
 - CI/CD usage, etc.

- Demonstration 10 minutes maximum

Part 4: Report

The report should **demonstrate critical reflection on the insights obtained from the interview discussion.**

- 8-10 pages
- Include workload distribution of the team members
- Page numbers must appear in the **footer**
- Cover Page (Mandatory)
Must include:
 - Course code and course name
 - Assignment title (Assignment 1)
 - Group number
 - Names and Registration numbers of all group members
 - Submission date

Deliverables

- Presentation slides – **This is for presentation purposes only. No need to submit.**
- Report – **Need to be submitted on or before the deadline.**
- Live demonstration – **Should be done during the viva slot.**
- LinkedIn profile link of the interviewed industry professional – **include in report**
- Video interview recording link – **Include in report.**
 - All **four group members must participate** in the interview
- Plagiarism report - The plagiarism should be less than 10% - **Include in report**

A mandatory viva will be conducted after the presentation and demonstration to assess individual understanding, practical knowledge, and active participation of all group members.

Marking Scheme

Criteria	Weight	Excellent (90-100%)	Good (75-89%)	Satisfactory (60-74%)	Poor (40 – 59%)	Very Poor (0-39%)
Interview Assesses the depth and relevance of insights from the industry professional, focusing on Quality Engineering across the SDLC.	25%	Deep, analytical discussion. Strong focus on SDLC integration, CI/CD, automation challenges, quality risks, and professional practices. Clear reflection beyond summary. All members actively participated.	Covers most required discussion areas. Good understanding of tool usage and QE practices. Some analytical reflection is present.	Interview completed, but mostly descriptive. Limited depth and analysis.	Superficial interview. Generic responses. Weak link to Quality Engineering.	Invalid interview, tool mismatch, missing evidence, or minimal engagement.
Report Assesses the ability to critically reflect on the interview insights and connect them with theoretical concepts from Software Engineering Process & Quality Management	25%	Strong critical reflection. Clear linkage between industry insights and Quality Engineering theory. Well-structured, professionally written, and meets all formatting requirements.	Good structure and reflection. Minor analytical or formatting weaknesses.	Mostly descriptive. Limited critical analysis. Some structural issues.	Weak organization and minimal reflection.	Incomplete or missing required components.
Presentation Assesses clarity, structure, justification of tool selection, and	15%	Professional slides, strong justification, accurate explanation of features and industry adoption.	Clear and structured presentation. Minor clarity or delivery issues.	Basic explanation. Limited depth or uneven participation.	Disorganized or unclear explanation.	Incomplete or weak presentation.

communication effectiveness.		Confident delivery.				
Demonstration Assesses practical implementation, correct tool usage, and feature demonstration in supporting Quality Engineering.	25%	Complete workflow demonstrated. Each member demonstrates at least one key feature (minimum four features per group). Clear explanation of quality benefits. Smooth, reliable execution.	Functional demo with required features implemented. Minor explanation gaps.	Demo works but lacks depth or full feature coverage.	Demo partially works. Limited understanding evident.	Demo is not functional or lacks required features.
Viva Assesses each member's individual understanding of the selected tool and related Quality Engineering concepts.	10%	Demonstrates strong conceptual and practical understanding. Confident and technically accurate responses.	Good understanding with minor gaps.	Basic understanding. Limited depth in responses.	Weak technical understanding.	Unable to explain tools, implementation, or concepts clearly.